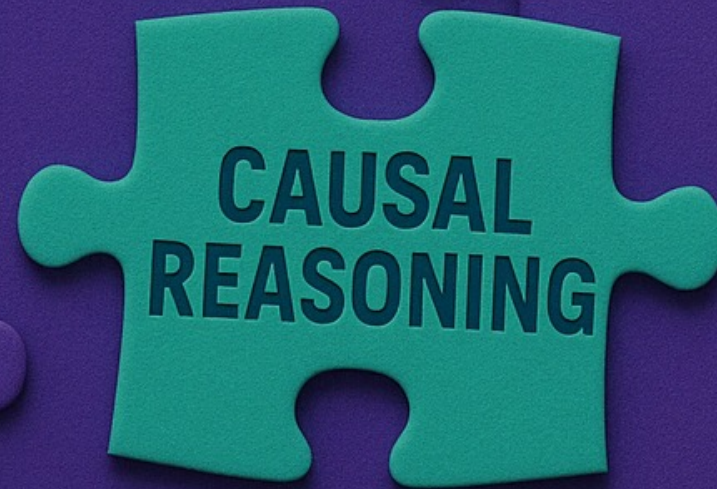


PART 1



ANNOUNCEMENT

- The prompt for your semester project is online. Please familiarize yourself with the prompt.
 - **Deliverable 1**: March 24th, 11:59 pm
 - **Deliverable 2**: April 5th, 11:59 pm
 - **Deliverable 3**: April 19th, 11:59 pm
 - **Deliverable 4**: May 1st, 11:59 pm

Deliverable 3

Presentation				
Criteria	> 85% Exemplary	75% – 85% Proficient	65% – 75% Developing	< 65% Needs Improvement
Research Question & Hypothesis (20%)	Question is novel, focused, clearly quantitative, and testable. Hypothesis is specific, directly stems from the question, and makes a clear, directional prediction.	Question is focused and testable. Hypothesis is clear and testable but may lack specificity or a strong theoretical link.	Question is too broad, vague, or not fully quantitative. Hypothesis is unclear, not directly testable, or does not logically follow from the question.	Question is descriptive, not explanatory, or is untestable. Hypothesis is missing or completely unrelated to the question.
Integration of Course Concepts (25%)	Proposal expertly uses and correctly defines course terms (e.g., IV/DV, confounding, operationalization, observation/intervention). Statistical work is perfectly matched to the hypothesis and variable types.	Proposal uses course terms correctly most of the time. Statistical work is appropriate, but justification may be slightly weak.	Use of course concepts is inconsistent or contains errors. Statistical work is questionable or poorly justified.	Fails to use or severely misuses core course concepts. Statistical work is missing or entirely inappropriate.
Methodological Rigor & Feasibility (25%)	Research design is the ideal choice for the question. Sampling plan, measures, and procedure are highly detailed, ethical, and clearly feasible. Limitations are thoughtfully addressed.	Research design is appropriate. Methods are clear and mostly feasible, though some details may be lacking.	Research design has major flaws for answering the question. Methods are vague, unethical, or clearly unfeasible. Key details are missing.	Methods section is incomplete, incoherent, or demonstrates a fundamental misunderstanding of research conduct.
Slide Design (15%)	Well-organized presentation with logical flow.	Generally well-structured but with minor clarity issues	Some structure but major clarity issues	Disorganized and unclear.
Communication (15%)	Well-paced, strong engagement, minimal reading	Some reliance on notes; pacing slightly uneven	Heavy reliance on reading, uneven pacing.	Difficult to follow; monotone; heavy reliance on reading

CAUSAL REASONING

- Today, we look into the prerequisites to the theory of causation:
 1. Observation vs. Intervention: The Difference
 2. Observation vs. Intervention: Predicting Depends Upon Which You Did and Upon What Causes What
 3. Events vs. Kinds of Events
 4. Particular Cases of Causation vs. Causal Generalizations
 5. The Idea of a Variable
 6. The Idea of a Data Table

CAUSAL REASONING

- **Causation** is best understood as how the world *responds to an intervention*.
 - Example: Boys who are watching the most violence on TV and getting into the most fist fights at school.
- **Association** does not imply **causation**.
 - Example: Association vs. Causation: Eye Color
 - Example: Association vs. Causation: Winter Coats and Air Temperature
- **Causal Reasoning** involves variables (e.g., weight, income); they are the foundation of data collection and statistical analyses.

CAUSAL REASONING

- (1) There is a difference between *observation* and *intervention*
 - Observation means passively doing something; intervention means actively changing something.
 - Note, passive observation sometimes involves acquiring data through measurement or other actions, e.g., asking questions. However, whatever action the observer takes, it does not change the thing being observed in a relevant way.
- Example: Observation vs. Intervention: Income

CAUSAL REASONING

- (2) Observation vs. Intervention: Predicting Depends Upon Which You Did and Upon What Causes What
 - What you can predict depends not only on whether you observed or intervened, but also on what is causing what.
 - Example: Gardens & Sprinklers

CAUSAL REASONING

- (3) Events vs. Kinds of Events: We distinguish between particular causal claims and causal generalization.
 - Particular causal claims involve one singular event (e.g., an accidental shooting at a party)
 - Causal generalizations involve general trends (e.g., a policy about school vouchers).
- Reasoning about particular causal claims depends upon our beliefs about causal generalization.
 - Example: Getting the flu

CAUSAL REASONING

- (4) Particular Cases of Causation vs. Causal Generalizations
 - Example: The Space Shuttle Columbia
 - Example: Cholesterol and Carbohydrates

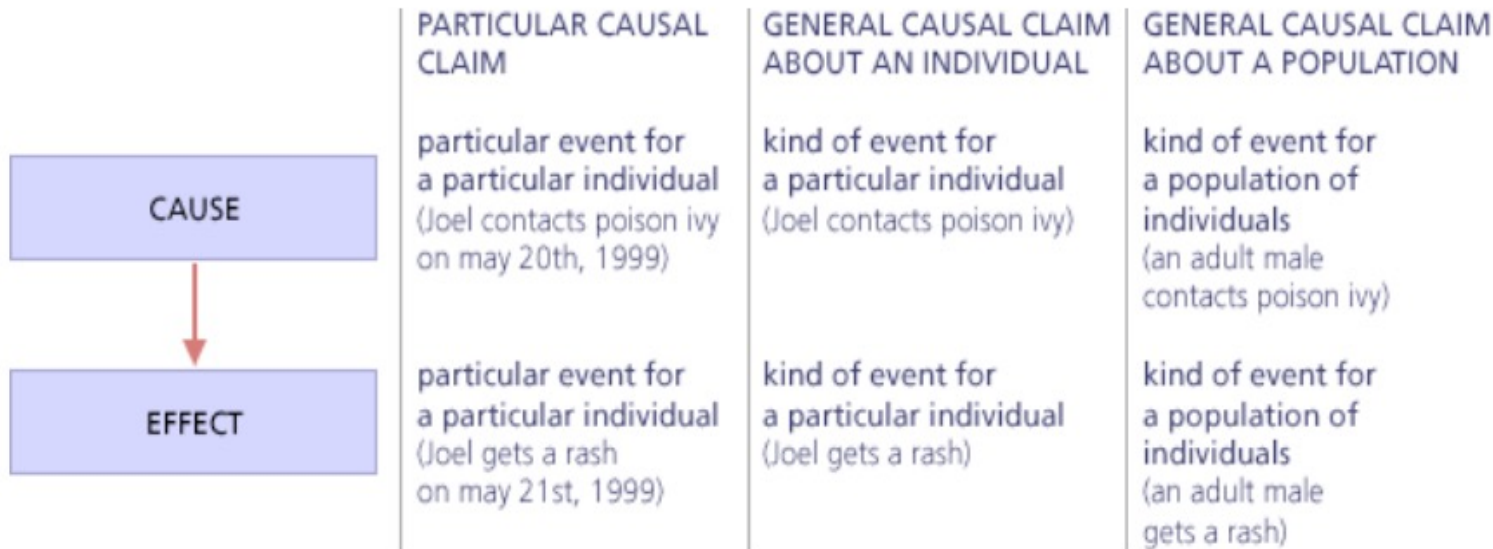
CAUSAL REASONING

- (5) The Idea of a Variable
 - Causal generalization: one *thing* is said to generally cause *another*.
 - Those “things” are our *variables*.

CAUSAL REASONING

- We group **events** and **conditions** and call them *factors*.
 - Conditions are enduring features of a situation, while events are changes in a situation.
 - A causal factor is called an event if it is an activity that changes a feature of the situation, e.g., setting the parking brake.
- Example of events in which one thing caused another:
 - The Titanic hits an iceberg and then sinks.
 - Pushing the gas pedal causes the car to accelerate.
- Example of conditions as causes or effects:
 - A street having a steep slope is a condition that causes a ball to roll down to the bottom of the hill.
 - The low density of oxygen at the top of the Mt. Everest is a condition that causes even moderate exercise to be exhausting.

CAUSAL REASONING



We generalize from particular causal events to general causal claims in two steps:

1. First, we generalize from a particular cause and effect event involving a single individual to ***events of a kind*** involving a single individual.
2. Then we move from single individuals to a ***group of individuals***.

CAUSAL REASONING

- (6) The Idea of a Data Table

Individual	Variable: Vaccine Received	Variable: Health Outcome
Lopez	live	OK
Mitchell	live	Polio
Shaffer	none	OK
Goodstein	dead	OK
Sufez	dead	OK

The columns in the table are variables: That is, quantities that can take on different **values** for different individuals.

The connection between events, conditions, kinds of events, and variables is this: **kinds of events (conditions)** can be represented as **values of variables**.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss - Seeing vs. Doing

 **Question No 01** – Consider the variables:

1. Pushing a shopping cart [yes, no]
2. Having sun-screen on [yes, no]
3. Having a coat on [yes, no]
4. Wearing a Red Bathing Suit [yes, no].

Which of the following are cases of **observing** the value of one of these variables as opposed to **intervening** to set their value? Select all that apply.

- A. At the supermarket, I noticed that everyone pushing a cart had a coat on.
- B. At the beach, I applied sun-screen to my forehead.
- C. At the supermarket, before I started shopping, I took my coat off, and I also took off the coat of my daughter.
- D. At the beach, three people walked by, and all of them had on red bathing suits.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss - Seeing vs. Doing

 **Question No 01** – Consider the variables:

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Which of the following are cases of observing the value of one of these variables as opposed to intervening to set their value? Select all that apply.

- A. At the supermarket, I noticed that everyone pushing a cart had a coat on. **[Observing]**
- B. At the beach, I applied sun-screen to my forehead. **[Intervening]**
- C. At the supermarket, before I started shopping, I took my coat off, and I also took off the coat of my daughter. **[Intervening]**
- D. At the beach, three people walked by, and all of them had on red bathing suits. **[Observing]**

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss - Seeing vs. Doing

Question No 02 – Suppose that a group of social researchers perform an experiment in which they assign different social routines to two groups of people. One group is required to gather twice a day to discuss current events and personal concerns, while the other group is required to see no one all day. **At the end of two months, the researchers measure the blood pressure levels of the people in the two groups. In this example, the researchers have: (choose one)**

- A. Passively observed the social routines people in the experiment.
- B. Passively observed both social routines and the blood pressure.
- C. Intervened to set the social routines and passively observed the blood pressure of the people in the experiment.
- D. Neither passively observed nor intervened.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss - Seeing vs. Doing

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- D. Neither passively observed nor intervened.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss - Predicting, Seeing, and Doing

Question No 03 – Suppose that you get to look at both of my next door neighbor's tax returns. Suppose you notice that the neighbor on my left paid \$17,000 in taxes, and the neighbor on my right paid \$44,000 in taxes. **Although you cannot know for sure, who do you predict has a higher income? (Select one answer.)**

- A. My neighbor who paid \$17,000 in taxes.
- B. My neighbor who paid \$44,000 in taxes.
- C. The information in the questions give no information about their income.

Reading Exercise – No 13 (in-class activity)

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Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss - Predicting, Seeing, and Doing

Question No 04 – Suppose you work for the IRS and you get to go into the IRS computer and specify how much tax both of my next door neighbors owe. Suppose you assign my neighbor on the left to pay \$34,000 in taxes, and the neighbor on my right \$24,000 in taxes. **Now who do you predict has a higher income? (Select one answer.)**

- A. My neighbor who you made owe a tax of \$34,000
- B. My neighbor who you made owe a tax of \$24,000
- C. The question gives no information about their income.

Reading Exercise – No 13 (in-class activity)

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Reading Exercise – No 13 (in-class activity)

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B - Reflect and discuss - Predicting, Seeing, and Doing

Question No 05 – Suppose you just opened a business and hired both of my next door neighbors. Suppose you assign my neighbor on the left a salary of \$64,000, and the neighbor on my right a salary of \$42,000. **Which neighbor do you predict will pay more in taxes? (Select one answer.)**

- A. My neighbor who you gave a salary of \$64,000
- B. My neighbor who you gave a salary of \$42,000
- C. The information in the questions give no information about their income.

Reading Exercise – No 13 (in-class activity)

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- C. The information in the questions give no information about their income.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss - Predicting, Seeing, and Doing

 **Question No 06** – Consider the following two cases:

1. You randomly select a person from the phone book and pay them to wear a knee brace.
2. You pass a person on the street who is wearing a knee brace.

In which of these cases can you reasonably predict that the person has an injured knee? (Select one answer.)

- A. Both 1 and 2
- B. 1 only
- C. 2 only
- D. Neither 1 nor 2

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss - Predicting, Seeing, and Doing

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Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - Particular Events and Causal Generalizations

 **Question No 07 – Which of the following are causal generalizations, as opposed to particular events involving causation? (Select all that apply.)**

- A. Playing violent video games leads to anti-social behavior.
- B. The U.S. entered World War II because of Pearl Harbor.
- C. The result of high cholesterol is heart disease.
- D. The Apollo 1 capsule fire was the result of an electrical short.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

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Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - Particular Events and Causal Generalizations

 Question No 08 – Which of the following is a causal generalization rather than a particular causal claim? (Select one answer.)

- A. Fred failed the exam because he got drunk the night before
- B. Yesterday, Jim hit the brakes that made him go into a skid.
- C. Playing violent video games causes children to behave aggressively
- D. Last week, Tommy hit his sister because he was in a bad mood from playing Doom for five hours.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - Particular Events and Causal Generalizations

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Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Events and Conditions I

 **Question No 09 – Which of the following causal factors are best classified as events rather than conditions? (Select all that apply.)**

- A. Hitting a home run.
- B. Living in the USA.
- C. Spraining your ankle.
- D. Being Hispanic.
- E. Liking rap music.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

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Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Events and Conditions I

 **Question No 10 – Which of the following causal factors are best classified as conditions rather than events? (Select all that apply.)**

- A. Owning a computer.
- B. Buying a computer.
- C. Getting a raise in salary.
- D. Being male.
- E. None of the above.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Events and Conditions I

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- E. None of the above.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss - Events and Conditions II

Question No 11 – Consider the factors affecting the movement of a (currently stationary) car. You can press the accelerator, apply the parking brake, and turn the lights on or off. Think about what events and conditions cause the car to move, and which keep it still. **The event of releasing the parking brake _____ results in the car moving. (Select one answer.)**

- A. Always
- B. Never
- C. Sometimes

Reading Exercise – No 13 (in-class activity)

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 **Question No 12** – Do the lights cause the car to move? (Select one answer.)

- A. Yes
- B. No
- C. No way to tell

Reading Exercise – No 13 (in-class activity)

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 **Question No 13** – Suppose that the following conditions are met: parking brake is off, lights are on. **What event will cause the car to move forward? (Select one answer.)**

- A. Pushing the accelerator down
- B. Letting the accelerator come up
- C. No event will cause it to move

Reading Exercise – No 13 (in-class activity)

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Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss - Events to kind of Events

 **Question No 14** – Referring to Table 1 in your reading, **which of the following is a kind of event for the individual, Bill Clinton. (Select one answer.)**

- A. An 11:00 A.M., May 7, 1992 fund raiser attended by Bill Clinton.
- B. A campaign for political office by Bill Clinton.
- C. A campaign for political office.
- D. A campaign for political office by George Bush.

Reading Exercise – No 13 (in-class activity)

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Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss - Events to kind of Events

Question No 15 – Refer again to the row in Table 1 in which the particular event is: Bill Clinton's campaign for political office in 1992.
Which of the following would be a kind of event generally? (Select one answer.)

- A. An 11:00 A.M., May 7, fund raised attended by Bill Clinton.
- B. A campaign for political office by Bill Clinton
- C. A campaign for political office.

Reading Exercise – No 13 (in-class activity)

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- B. A campaign for political office by Bill Clinton
- C. A campaign for political office.**

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss - Events to kind of Events

 **Question No 16** – Refer to the row in the Table 1 for the event - The U.S. Stock Market crash of October 17, 1987. **Which of the following is a kind of event for the individual? (Select one answer.)**

- A. The Tokyo Stock Market crash of 1987
- B. U.S. Stock Market crashes
- C. 1987 U.S. Stock Market recovery

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss - Events to kind of Events

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Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss - Events to kind of Events

Question No 17 – Refer to the row in the Table 1 for the event - The U.S. Stock Market crash October 17, 1987. **Which of the following is a kind of event generally? (Select one answer.)**

- A. Stock market crashes
- B. The Tokyo Stock Market crash of 1987
- C. The U.S. Stock Market crash of 1987.

Reading Exercise – No 13 (in-class activity)

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Reading Exercise – No 13 (in-class activity)

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G - Reflect and discuss - Variables and Data Tables 1

Question No 18 – It is known that being male is a cause of having a heart attack. Thus, men have a higher frequency of heart attacks than do women. **What are the variables that are causes referred to in this claim? (Select all that apply.)**

- A. Number of heart attacks.
- B. Gender.
- C. Number of heart attacks for men.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

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Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss - Variables and Data Tables 1

Question No 19 – A drunk driver with malfunctioning anti-lock brakes on a wet road hits a car in front that came to a sudden stop. **Which of the following uses variables and values to properly represent the cause event in this situation. (Select one answer.)**

- A. Car in front=sudden stop
- B. Car in back=drunk driver, malfunctioning brakes, wet road, sudden stop
- C. Front car makes sudden stop=true
- D. Drunk=true, Anti-lock Brakes=malfunctioning, Road Condition=Wet, Sudden stop = true

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

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- D. Drunk=true, Anti-lock Brakes=malfunctioning, Road Condition=Wet, Sudden stop = true**

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss - Variables and Data Tables 1

Question No 20 – The more someone smokes, the more likely he or she is to develop emphysema. **Which of the following might serve as a variable to represent the cause in this case? (Check all that apply)**

- A. Number of cigarettes smoked per day.
- B. Lung capacity in liters.
- C. Number of packs of cigarettes smoked per year.
- D. Heart rate of the person during a stress test.

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss - Variables and Data Tables 1

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Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss - Variables and Data Tables 2

 **Question No 21** – Consider the following passage:

A recent study has found that, among women above the age of 65, there is a strong link between exercise and bone density. Among the women in the study who did not exercise regularly, twenty-five percent had osteoporosis (dangerously thin bones), while this number dropped to less than 10 percent among those who did exercise on a regular basis. Lead researcher Julie Ripker stated that, “These data further demonstrate the beneficial effects of exercise upon quality of life for senior citizens.” **Which of the following is the variable being considered as the cause in the passage? (Select one answer.)**

- A. Regular Exercise [yes, no]
- B. Age
- C. No Exercise = 25%
- D. Osteoporosis [yes, no]

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss - Variables and Data Tables 2

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- A. Regular Exercise [yes, no]
- B. Age
- C. No Exercise = 25%
- D. Osteoporosis [yes, no]

 **Question No 22** – Consider the same passage as above (No 21). **Which of the following is the variable being considered as the effect in the passage? (Select one answer.)**

- A. Regular Exercise [yes, no]
- B. Age
- C. No Exercise = 25%
- D. Osteoporosis [yes, no]

Reading Exercise – No 13 (in-class activity)

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H - Reflect and discuss - Variables and Data Tables 2

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- C. No Exercise = 25%
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- A. Regular Exercise [yes, no]
- B. Age
- C. No Exercise = 25%
- D. Osteoporosis [yes, no]**

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss - Study: Cell phones not linked to cancer

Read the case study below on cell phones and cancer, and then answer the questions that follow.

Date: December 20, 2000 Source : <http://www.usatoday.com> Copyright: 2001 USA TODAY

Concepts -

- variables
- causal graphs

Keywords -

- cell phones
- cancer

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss - Study: Cell phones not linked to cancer

Scientists have good news for cellphone junkies: The ubiquitous gadgets, used by more than 90 million Americans, don't appear to raise the risk of brain cancer, says a study released today. Joshua Muscat of the American Health Foundation in Valhalla, N.Y., and colleagues report in today's Journal of the American Medical Association that there is no significant risk from short-term use of cellular phones that use analog signals.

But this study will by no means settle the debate. Muscat and colleagues caution that more research is needed to determine health effects of long-term phone use.

The research, funded by federal and cellphone industry grants, compared cellphone habits of 469 people with brain cancer to those of 422 healthy people matched by age, sex and other characteristics.

Study participants in the group with cancer averaged 2.5 hours of phone use per month, while the control group averaged 2.2 hours per month.

Except for preliminary results from a Swedish study, which also found no risk from the phones, there have been no studies on how cellphones affect humans, Muscat says. "No one study is ever definitive," he says, "but if these results are confirmed in other studies which are ongoing in the U.S. and Europe, we have the evidence to show the phones can be used with confidence, without there being a cancer risk."

But Louis Slesin, editor of Microwave News, a New York-based newsletter on electromagnetic radiation and health, criticizes the study as offering "answers to last year's questions" by looking only at use of analog phones.

"The industry has now moved to digital," he says, and "some of the research shown in animals has found no effect in analog but some concern in digital. The technology is moving much faster than the health research, so we're always one step behind."

Muscat says studies are underway in Europe to investigate the possible risks of digital cellphones. In the journal article, the researchers say these newer phones produce signals that "might have different effects on biological tissues than analog telephones."

Slesin says the new report "is welcome news, and we don't wish anyone ill," but "my concern is people will rely on the study too much as an all-clear, when it's really the beginning of the story, not the end."

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss - Study: Cell phones not linked to cancer

Question No 23 – In the article on Cell Phones, which of the following are variables with an appropriate range of values that are discussed in the article as effects? (Select all that apply.)

- A. Cell Phone Use [none, low, high].
- B. Cell Phone Use [0... 100 hours per month].
- C. Have Brain Cancer [yes, no].
- D. Have Cancer [none, lung cancer, brain cancer].

Reading Exercise – No 13 (in-class activity)

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Reading Exercise – No 13 (in-class activity)

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I - Reflect and discuss - Study: Cell phones not linked to cancer

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- C. Have Brain Cancer [yes, no].
- D. Have Cancer [none, lung cancer, brain cancer].



Question No 24 – In the article on Cell Phones, which of the following are variables with an appropriate range of values that are discussed in the article as the cause of interest? (Select one answer.)

- A. Age [in years]
- B. Sex [male, female]
- C. Cell Phone Use [0..100 hours per month]
- D. Have Brain Cancer [yes, no]

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss - Study: Cell phones not linked to cancer

Question No 23 – In the article on Cell Phones, which of the following are variables with an appropriate range of values that are discussed in the article as effects? (Select all that apply.)

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- B. Cell Phone Use [0... 100 hours per month].
- C. Have Brain Cancer [yes, no].**
- D. Have Cancer [none, lung cancer, brain cancer].

 **Question No 24 – In the article on Cell Phones, which of the following are variables with an appropriate range of values that are discussed in the article as the cause of interest? (Select one answer.)**

- A. Age [in years]
- B. Sex [male, female]
- C. Cell Phone Use [0..100 hours per month]**
- D. Have Brain Cancer [yes, no]

Reading Exercise – No 13 (in-class activity)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss - Link it to your research proposal



Many of the ideas you have learned in this handout will be relevant to your Semester Research Project. **Discuss the connection of these ideas to your project here in as much detail as possible, seeking help from your instructor when needed.** (Refer back to Barroga and Matanguihan (2022) for more help. If your project has not progressed to this stage, you should come back to this activity later.